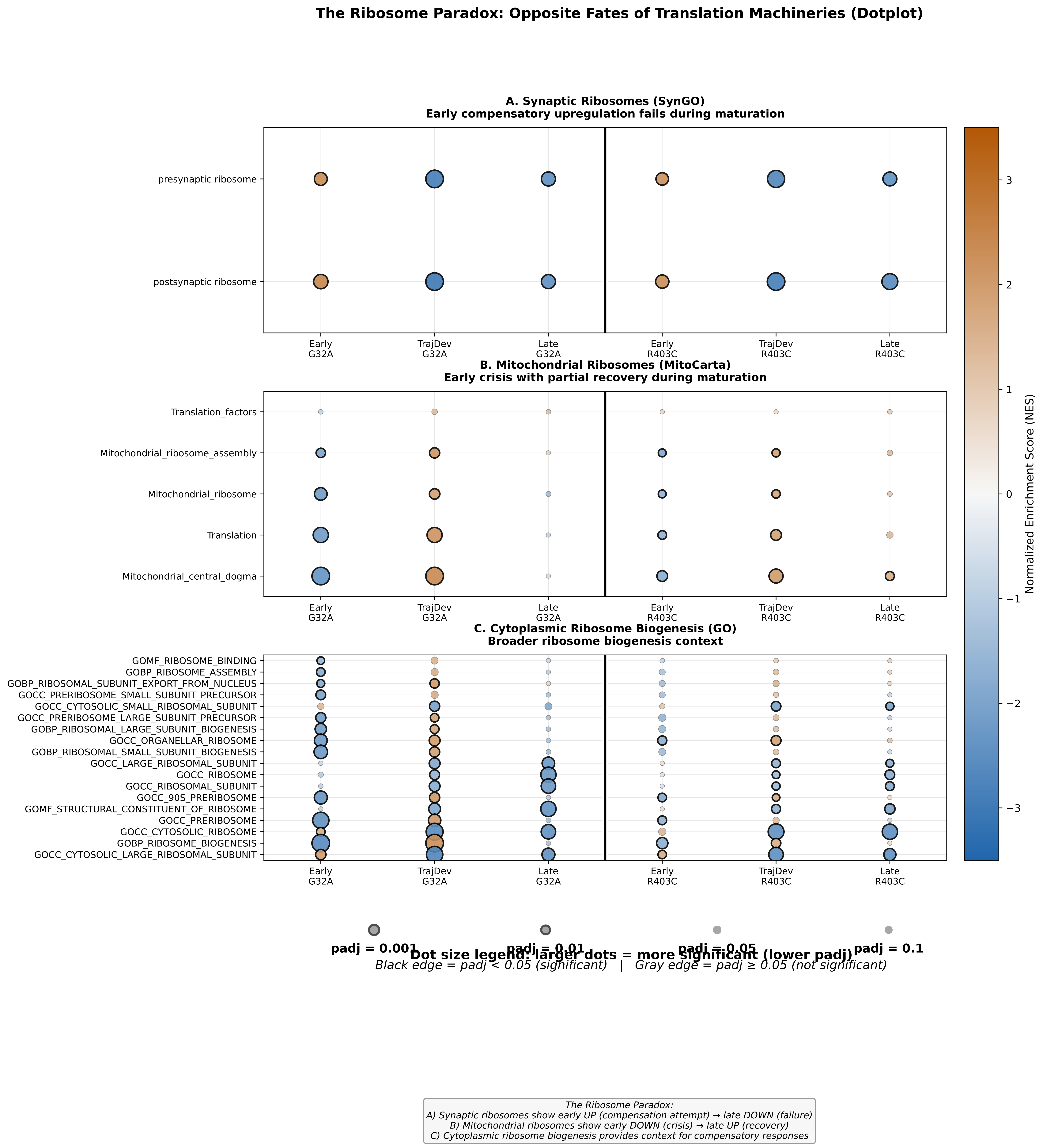

The Ribosome Paradox: Opposite Fates of Translation Machineries (Dotplot)
A. Synaptic Ribosomes (SynGO)
Early compensatory upregulation fails during maturation
presynaptic ribosome
postsynaptic ribosome
B. Mitochondrial Ribosomes (MitoCarta)
Early crisis with partial recovery during maturation
Translation_factors
Mitochondrial_ribosome_assembly
Mitochondrial_ribosome
Translation
Mitochondrial_central_dogma
C. Cytoplasmic Ribosome Biogenesis (GO)
Broader ribosome biogenesis context
GOMF_RIBOSOME_BINDING
GOBP_RIBOSOME_ASSEMBLY
GOBP_RIBOSOMAL_SUBUNIT_EXPORT_FROM_NUCLEUS
GOCC_PRERIBOSOME_SMALL_SUBUNIT_PRECURSOR
GOCC_CYTOSOLIC_SMALL_RIBOSOMAL_SUBUNIT
GOCC_PRERIBOSOME_LARGE_SUBUNIT_PRECURSOR
GOBP_RIBOSOMAL_LARGE_SUBUNIT_BIOGENESIS
GOCC_ORGANELLAR_RIBOSOME
GOBP_RIBOSOMAL_SMALL_SUBUNIT_BIOGENESIS
GOCC_LARGE_RIBOSOMAL_SUBUNIT
GOCC_RIBOSOME
GOCC_RIBOSOMAL_SUBUNIT
GOCC_90S_PRERIBOSOME
GOMF_STRUCTURAL_CONSTITUENT_OF_RIBOSOME
GOCC_PRERIBOSOME
GOCC_CYTOSOLIC_RIBOSOME
GOBP_RIBOSOME_BIOGENESIS
GOCC_CYTOSOLIC_LARGE_RIBOSOMAL_SUBUNIT
Early G32A
TrajDev G32A
Late G32A
Early R403C
TrajDev R403C
Late R403C
Normalized Enrichment Score (NES)
3
2
1
0
−1
−2
−3
padj = 0.001
padj = 0.01
padj = 0.05
padj = 0.1
Dot size legend: larger dots = more significant (lower padj)
Black edge = padj < 0.05 (significant) | Gray edge = padj ≥ 0.05 (not significant)
The Ribosome Paradox:
A) Synaptic ribosomes show early UP (compensation attempt) → late DOWN (failure)
B) Mitochondrial ribosomes show early DOWN (crisis) → late UP (recovery)
C) Cytoplasmic ribosome biogenesis provides context for compensatory responses